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Master Thesis

Randomized Identification Codes with a Focus on K-Identification

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Topic: Randomized Identification Codes with a Focus on K-Identification

Task description:

Identification via channels (ID) asks receivers to decide whether a specific message was sent, rather than to reconstruct it. Randomized ID codes achieve doubly-exponential message sets in blocklength and enable vanishing Type-I (false-accept) and Type-II (miss) error probabilities at positive ID rate. K-Identification (K-ID) generalizes this: the receiver asks whether the transmitted message belongs to a set of K candidates. K-ID is relevant for goal/semantics-driven communication (e.g., digital twins), where the action depends only on whether the current state falls into a small set of actionable hypotheses, not on exact message recovery. This thesis will study randomized ID codes with emphasis on K-ID, and extend the ideas and evaluation from “Stopping the Data Flood: Post-Shannon traffic reduction in digital-twins applications” to the K-ID setting. The work should balance theory, algorithms, and software-driven experiments using our provided library.

The student is expected to:

Survey the state of the art:

- Core ID concepts: randomized encoders/acceptance regions, ID capacity, error exponents, and operational interpretations; contrast with Shannon transmission.
- Define K-ID precisely (membership in a K-set) and position it vs. related notions (multiple-object ID, generalized ID; deterministic vs randomized variants).

Formulate the problem & metrics for K-ID:

- Formal model (channel class, randomization/common randomness budget, decoder tests).
- Report both Type-I/Type-II errors and computational metrics (encode/decode time, seed length, memory), plus finite-blocklength behavior.

Replicate & extend the digital-twins baseline to K-ID:

- Reproduce key experiments/results from the “Stopping the Data Flood” study (baseline ID) with our software stack.
- Design K-ID experiments that reflect digital-twin quantifying traffic reduction vs. reliability/latency.

Use our software library to build a modular K-ID test bench:

- Implement randomized ID/K-ID encoders and decoders (acceptance region tests) as pluggable components.

Design and run a simulation campaign:

- Measure KPIs such as: Type-I/II error curves, empirical exponents, queries per symbol, and compute/memory usage. Quantify traffic reduction vs. reliability/latency

Deliver a comparative assessment with clear takeaways:

- When do systems benefit from K-ID?
- Tradeoffs among randomization, complexity, and reliability.

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Statement of authorship

I hereby certify that I have authored this document entitled *Randomized Identification Codes with a Focus on K-Identification* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. During the preparation of this document I was only supported by the following persons:

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Additional persons were not involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

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Abstract

Identification via channels (ID) asks a receiver to decide whether a specific hypothesis about the transmitted message is true, rather than reconstructing the message itself. This document provides precise definitions, key capacity results, error metrics, and an experimental plan tailored to K-Identification (K-ID), where the receiver asks whether the message belongs to a set of K candidates. It contrasts randomized vs. deterministic (DI) variants, formulates problem and metrics, and outlines a modular test bench and simulation campaign aligned with the provided software stack.

Contents

List of Figures	III
List of Tables	IV
Acronyms	V
1 Introduction	1
2 Background and Related Work	3
2.1 From Data Reconstruction to Semantic Identification	3
2.2 Alternative Synchronization Strategies	4
2.2.1 Full State Transmission (The Baseline)	4
2.2.2 Delta Encoding and the Drift Problem	4
2.2.3 Event-Triggered Control (ETC)	4
2.2.4 The Identification Advantage	5
2.3 Encoding Mechanisms: Mathematical Structure and Bias	5
2.3.1 Cryptographic Construction: The SHA-256 Architecture	5
2.3.2 Algebraic Construction: The Geometry of RS-ID	6
2.3.3 Structural Blindness and Randomization	7
2.4 Operational Prerequisite: Common Randomness Generation	7
2.5 Theoretical Limits and Algorithmic Trade-offs	8
2.5.1 The Algebraic Limit: Saturation and Field Size	8
2.5.2 The Cryptographic Perspective: Independence from Data Size	9
2.5.3 Engineering Trade-offs: Bound vs. Speed	9
2.6 Control-Theoretic Perspectives: Deadbands and Stability	9
2.6.1 K-ID as Deadband Control	9
2.6.2 Prevention of Zeno Behavior	10
2.6.3 Bounded-Input Bounded-Output (BIBO) Stability	10
2.7 Semantic Verification: Adapting to Control Flexibility	10
2.7.1 The Autonomous Driving Scenario	10
2.7.2 Human-Robot Collaboration and Jitter	11
2.7.3 Discussion: Trading Freshness for Value	11
3 Design Methodology	12
3.1 System Model and Data Flow	12
3.1.1 Baseline: Traditional Transmission Flow	12
3.1.2 Proposed: Identification (ID) Data Flow	13

3.2	Encoding Mechanisms and Reliability Analysis	13
3.2.1	Algebraic Implementation and Error Bound (RS-ID)	13
3.2.2	Cryptographic Construction and Error Analysis (SHA-256)	14
3.2.3	Tolerance-Based Verification (K-ID)	15
3.3	End-to-End Latency Model	16
4	Implementation	17
4.1	Experimental Parameters	17
4.2	Theoretical Collision Probabilities	17
4.3	Latency Plot Generation	17
4.3.1	Parameters	17
4.3.2	Logic	17
4.3.3	SHA-256	18
4.3.4	RS-ID	18
4.4	K-ID Parameter Sweep Setup	18
4.4.1	Model and Metrics	18
4.4.2	State Space and Quantization	19
4.4.3	Parameter Grid	19
4.4.4	Reproducibility	19
5	Results and Evaluation	20
5.1	Empirical Parameter Verification	20
5.1.1	Encoding Latency (t_{ver})	20
5.1.2	Theoretical Collision Probability (p_{miss})	21
5.2	End-to-End Latency Evaluation	22
5.2.1	Sensitivity to Channel Bandwidth	22
5.2.2	Sensitivity to Prediction Quality	22
5.3	K-ID Safety–Latency Trade-off	24
5.3.1	Sweep Results	24
5.3.2	Pareto Frontier Interpretation	24
5.3.3	RS-ID Comparison and Uniform Error Model	27
6	Conclusion and Future Work	29
6.1	Conclusion	29
6.2	Future Work	30
6.2.1	Explicit Common Randomness Synchronization	30
6.2.2	Semantic Set Design Beyond Scalar Deadbands	30
6.2.3	Adaptive Policies for (K, n_{ver})	31
6.2.4	Non-Ideal Networks and Protocol Mechanics	31
6.2.5	Implementation Efficiency and Hardware Realism	31
6.2.6	Broader Threat Models and Robustness	31
	Bibliography	32
A	Appendix	34

List of Figures

3.1	Data flow of Traditional Synchronization: The full state is transmitted regardless of the Digital Twin's knowledge.	12
3.2	Data flow of ID Synchronization: High-bandwidth transmission is replaced by local encoding and conditional fallback.	13
5.3	K-ID parameter sweep results for $n_{data} = 96$ bits at $B = 1$ Mbps.	25
5.4	Pareto view of the K-ID trade-off: speedup vs. conditional miss rate $P(\text{accept} \mid \text{unsafe})$	26
5.5	Pareto view using overall risk $P(\text{unsafe} \cap \text{accept})$: speedup vs. total unsafe-accept probability per tick.	26
5.6	K-ID sweep comparison of truncated SHA-256 vs. RS-ID under a uniform prediction error model ($n_{data} = 96$ bits, $B = 1$ Mbps).	28

List of Tables

5.1	Measured Verification Time t_{ver} (mean $\pm$ 95% CI half-width, with relative half-width)	21
5.2	Theoretical Collision Probability (p_{miss}) for Bit-Based Parameters	21

Acronyms

AoI	Age of Information	11
BIBO	Bounded-Input Bounded-Output	10
CPS	Cyber-Physical Systems	1
CR	Common Randomness	7
DRBG	Deterministic Random Bit Generator	8
DT	Digital Twin	1–5, 7, 8, 10, 11
ETC	Event-Triggered Control	4, 5, 10
HRC	Human-Robot Collaboration	11
ID	Identification	1–3, 5, 7, 10, 11
K-ID	K-Identification	1–3, 9–11
NCS	Networked Control System	9
PLKG	Physical Layer Key Generation	8
PRNG	Pseudo-Random Number Generator	8
PTP	Precision Time Protocol	8
RS-ID	Reed-Solomon Identification	2, 6–9
SHA-256	Secure Hash Algorithm 256-bit	2, 7, 9
TCP	Transmission Control Protocol	4
UDP	User Datagram Protocol	4
URLLC	Ultra-Reliable Low-Latency Communications	1, 3, 4
Vol	Value of Information	11

1 Introduction

Digital Twins (DTs) have become a standard component in modern industrial automation for connecting physical assets with their virtual counterparts. By maintaining accurate virtual copies of systems like industrial robots, DTs enable real-time monitoring, predictive maintenance, and remote control. In time-critical Cyber-Physical Systems (CPS) applications, the utility of a DT is strictly defined by the synchronization accuracy between the physical robot and the digital model [19]. Consequently, ensuring state consistency typically demands Ultra-Reliable Low-Latency Communications (URLLC) capabilities to support high-frequency updates.

However, meeting these requirements via standard periodic transmission is resource-intensive. Since the DT often possesses a precise estimate of the robot’s state, continuously transmitting raw telemetry data creates significant redundancy. This inefficiency consumes scarce bandwidth without providing new information gain.

To mitigate this redundancy, recent research proposes *identification-based Identification (ID)* synchronization [12]. Unlike classical Shannon-type communication, which aims to reconstruct the full message, ID schemes focus on a verification task: confirming whether the remote state matches a local hypothesis. Building on this concept, prior works [15] demonstrate that transmitting a compressed “verifier” (e.g., a short hash or randomized code) instead of the full state significantly reduces communication traffic. This approach effectively supports goal-oriented communication [6] in low-latency scenarios.

However, implementing ID synchronization in real-time control loops reveals two critical engineering trade-offs. First, regarding latency, the computational overhead of encoding on edge devices may exceed the time saved by reducing transmission traffic. This shift from a communication-bound to a computation-bound bottleneck necessitates a rigorous end-to-end feasibility analysis. Second, reliability becomes a primary concern when moving from exact to semantic verification. In practical control, enforcing bitwise synchronization (standard ID) is often overly conservative, triggering unnecessary updates for benign deviations such as sensor noise. To address this, we extend identification to semantic “safe ranges”, where the system verifies set membership rather than a unique value. This relaxation, however, introduces collision risks, specifically Type II (miss) errors, where dangerous desynchronizations are incorrectly classified as safe. Therefore, quantifying the relationship between the semantic acceptance range and safety risks is essential.

To address these limitations, this thesis presents a complementary study to existing ID frameworks, extending the analysis from pure traffic optimization to an integrated view of latency and reliability in K-Identification (K-ID) systems. The main contributions are:

First, we develop an end-to-end latency model combining computational and transmission delays. We compare cryptographic ([Secure Hash Algorithm 256-bit \(SHA-256\)](#)) and algebraic ([Reed-Solomon Identification \(RS-ID\)](#)) encoding schemes to determine the quantitative thresholds (break-even points) where [ID](#) yields a net latency benefit as a function of bandwidth and verifier length.

Second, we formalize the [K-ID](#) problem for robotic monitoring, distinguishing *safe* from *unsafe* desynchronization via semantic safety sets. We derive quantitative bounds on miss error probabilities induced by [K-ID](#) mechanisms.

Third, we conduct a systematic evaluation using a reproducible software stack. Through Monte Carlo experiments on Gaussian robot trajectories, we expose a non-monotonic dependence between the semantic acceptance range (Set B) and error rates. These results provide design guidelines for balancing semantic tolerance with safety detection in edge-based [DTs](#) networks.

Relation to prior work. While [ID](#) rests on a solid information-theoretic foundation, engineering studies have primarily focused on traffic suppression. We extend this line of work by treating latency as a primary metric alongside traffic and by generalizing from single identification to [K-ID](#), which aligns better with real-world [DT](#) workflows. Our results aim to inform design choices regarding *when to use ID/K-ID* and *with which verifier lengths* under realistic bandwidth conditions.

Organization. Chapter [2](#) introduces the background and related work; Chapter [3](#) details the system model and measurement protocol; Chapter [4](#) describes the software stack and reproducibility; Chapter [5](#) presents the experimental results, including latency–traffic curves parameterized by bandwidth and load; Chapter [6](#) concludes and outlines future engineering directions for [K-ID](#).

2 Background and Related Work

This chapter reviews the transition from classical data transmission to identification-based synchronization, positioning it within the broader context of semantic communication [20] for 6G networks. We analyze alternative synchronization strategies—such as differential encoding and event-triggered control—to motivate the engineering need for the identification (ID) paradigm. We then compare practical encoding mechanisms by contrasting the avalanche dynamics of cryptographic hashing with the geometric properties of algebraic coding. Finally, we discuss how this paradigm generalizes to K-ID, where strict equality constraints are relaxed to semantic safety sets to accommodate natural variability in physical control loops.

2.1 From Data Reconstruction to Semantic Identification

In time-critical control loops, the utility of a DT depends on its synchronization with the physical asset. Conventional protocols treat this as a data reconstruction problem, where the robot periodically transmits its full state vector S_t (e.g., joint angles and velocities) to the DT. While robust, this approach overlooks a key redundancy: the DT, using its internal kinematic model, already maintains a prediction $\hat{S}_t$ of the robot’s state. The communication task is therefore not to inform the DT “where the robot is,” but to verify “whether the DT’s prediction is correct.”

This shift aligns with the paradigm of **semantic communication** [17] in 6G networks. Unlike classical Shannon-type communication, which emphasizes accurate reconstruction, semantic communication emphasizes information that matters for the receiver’s goal. In the context of URLLC, this suggests prioritizing “actionable” information—such as safety violations—over raw telemetry.

Accordingly, the coding objective shifts from Shannon’s transmission capacity to the identification capacity defined by Ahlswede and Dueck [2]. In transmission, each possible state must map to a disjoint decoding set, limiting the number of distinguishable states to 2^{nR} for blocklength n . In contrast, identification permits overlaps between decoding sets, as long as pairwise intersections are statistically negligible. This relaxation enables a doubly exponential growth in identifiable states ($2^{2^{nR}}$), meaning that short verifiers can discriminate among an extremely large state space while allowing a vanishingly small error probability.

2.2 Alternative Synchronization Strategies

Before detailing the implementation of identification, it is necessary to rigorously evaluate why established data reduction strategies—specifically Delta Encoding and Event-Triggered Control—are often insufficient for unreliable edge networks.

2.2.1 Full State Transmission (The Baseline)

The standard baseline in industrial automation is the periodic transmission of the full state vector S_t . This approach is favored for its statelessness; each packet is self-contained. If a packet at time t is lost due to channel interference, the packet at $t + 1$ allows the DT to immediately recover the correct state. However, this robustness comes at the cost of extreme bandwidth inefficiency. In steady-state operations, where $S_t \approx S_{t-1}$, the information entropy of the transmitted packet is near zero, yet it consumes the full bandwidth allocated to the payload n_{data} .

2.2.2 Delta Encoding and the Drift Problem

To address this redundancy, Delta Encoding [14] (or differential transmission) is frequently employed. This method transmits only the difference vector $\Delta_t = S_t - S_{t-1}$. Since Δ_t typically has a smaller dynamic range than the absolute state S_t , it can be quantized with fewer bits, offering significant compression ratios.

However, Delta Encoding introduces a key vulnerability: **statefulness**. The reconstruction of the current state at the receiver, denoted as S'_t , depends recursively on the history of received updates: $S'_t = S'_{t-1} + \Delta_t$. This recursive dependency creates a clear failure mode in unreliable transport protocols like User Datagram Protocol (UDP). A single lost packet Δ_k causes a persistent offset between the physical robot and the DT, a phenomenon known as “state drift.” [10] While reliable protocols like Transmission Control Protocol (TCP) can prevent packet loss, retransmissions introduce non-deterministic latency jitter, which can violate the timing deadlines of real-time control loops. Therefore, Delta Encoding poses a high risk for URLLC applications.

2.2.3 Event-Triggered Control (ETC)

Another prominent alternative is Event-Triggered Control (Event-Triggered Control (ETC)). Unlike Time-Triggered (periodic) systems, ETC only transmits data when the deviation between the current state and the last transmitted state exceeds a threshold δ : $\|S_t - S_{last_sent}\| > \delta$ [18]. This approach effectively silences the network during periods of inactivity.

Nevertheless, ETC relies fundamentally on a “push” communication model. A silent channel is ambiguous to the receiver: it could indicate that the system is operating safely within bounds, or it could indicate that the sensor has failed or the link has been severed. In safety-critical robotics, such ambiguity is often unacceptable. Furthermore, ETC requires the sensing node to perform continuous monitoring and threshold checking, pushing computational complexity to the energy-constrained edge [7].

2.2.4 The Identification Advantage

In contrast to these alternatives, the **ID** scheme proposed in this thesis combines statelessness with strong compression. Unlike Delta Encoding, **ID** verification checks the absolute state S_t directly and is therefore immune to drift; if a packet is lost, the next packet again provides an independent check. Unlike **ETC**, **ID** supports a “verify” model, allowing the **DT** to actively confirm synchronization with minimal overhead. This combination is well-suited for maintaining high-fidelity **DTs** over unreliable networks.

2.3 Encoding Mechanisms: Mathematical Structure and Bias

To realize the theoretical capacity of identification in edge computing, the encoding process must map a high-dimensional state space $\mathcal{S}$ into a low-dimensional verifier space $\mathcal{V}$ while minimizing collision probabilities. This section dissects the internal mechanisms of the two candidate algorithms—cryptographic hashing (SHA-256) and algebraic coding (RS-ID)—to reveal the mathematical origins of their reliability.

2.3.1 Cryptographic Construction: The SHA-256 Architecture

To understand the source of the high entropy in the generated verifiers, it is necessary to examine the internal structure of the SHA-256 algorithm. SHA-256 belongs to the SHA-2 family and is built upon the Merkle-Damgård construction, which iteratively processes the input state S using a dedicated compression function. The process consists of three mathematically distinct stages: preprocessing, message expansion, and the state update loop.

Preprocessing and Padding

The input message M (comprising the state S concatenated with the seed r) is first padded to ensure its length is a multiple of 512 bits. The padding scheme appends a single ‘1’ bit, followed by k zero bits, and finally a 64-bit integer representing the original message length. This padded message is parsed into N blocks of 512 bits, denoted as $M^{(1)}, M^{(2)}, \dots, M^{(N)}$.

Message Schedule (Expansion)

The core of the diffusion process begins with the Message Schedule. Each 512-bit block $M^{(i)}$ is expanded from 16 words (32-bit each) into 64 words, denoted as $W_0, \dots, W_{63}$. The first 16 words are copied directly from the message block. The remaining 48 words are generated recursively, creating a complex dependency chain where every future bit depends on multiple past bits:

$$W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \pmod{2^{32}} \quad (2.1)$$

where the small sigma functions σ_0 and σ_1 introduce the initial bitwise diffusion via rotation (ROTR) and shifting (SHR):

$$\sigma_0(x) = \text{ROTR}^7(x) \oplus \text{ROTR}^{18}(x) \oplus \text{SHR}^3(x) \quad (2.2)$$

$$\sigma_1(x) = \text{ROTR}^{17}(x) \oplus \text{ROTR}^{19}(x) \oplus \text{SHR}^{10}(x) \quad (2.3)$$

This expansion ensures that a single bit flip in the input $M^{(i)}$ propagates to affect a vast majority of the 64 words in the schedule, setting the stage for the avalanche effect.

The Compression Loop

The 64 expanded words W_t are then fed into the compression function. The working state consists of eight 32-bit variables, labeled a through h , initialized to constant prime-root values. For each step t from 0 to 63, the state is updated using two temporary variables T_1 and T_2 :

$$T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t \quad (2.4)$$

$$T_2 = \Sigma_0(a) + Maj(a, b, c) \quad (2.5)$$

Here, K_t represents 64 constant 32-bit words. The non-linear logical functions Ch (Choose) and Maj (Majority), along with the big sigma functions Σ_0 and Σ_1 , serve as the mixing engine:

$$Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z) \quad (2.6)$$

$$Maj(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z) \quad (2.7)$$

$$\Sigma_0(x) = ROTR^2(x) \oplus ROTR^{13}(x) \oplus ROTR^{22}(x) \quad (2.8)$$

$$\Sigma_1(x) = ROTR^6(x) \oplus ROTR^{11}(x) \oplus ROTR^{25}(x) \quad (2.9)$$

After calculating T_1 and T_2 , the registers are shifted ($h \leftarrow g, g \leftarrow f, \dots$), and updated as:

$$e \leftarrow d + T_1, \quad a \leftarrow T_1 + T_2 \quad (2.10)$$

This rigorous mixing guarantees that after 64 rounds, every bit of the final hash output is a complex, non-linear function of every bit of the input block. Finally, the output verifier v is obtained by truncating this digest to the target length n_{ver} , retaining the high-entropy properties generated by this procedure.

2.3.2 Algebraic Construction: The Geometry of RS-ID

In contrast to the bitwise diffusion of SHA-256, [RS-ID](#) relies on the geometric properties of polynomials over finite fields.

Data as a Polynomial Curve

Let the state vector S be decomposed into k segments $\{s_0, s_1, \dots, s_{k-1}\}$, where each segment is an element of the Galois Field $\mathbb{F}_q$. We interpret these segments not as raw data, but as coefficients of a unique polynomial curve $P_S(x)$ of degree $k - 1$:

$$P_S(x) = s_0 + s_1x + s_2x^2 + \dots + s_{k-1}x^{k-1} = \sum_{i=0}^{k-1} s_i x^i \quad (2.11)$$

The generation of a verifier v using a random seed r is geometrically equivalent to **sampling this curve** at the coordinate $x = r$:

$$v = P_S(r) \pmod{q} \quad (2.12)$$

Collision as Intersection

A collision (Type II error) occurs when the true state S and the predicted state $\hat{S}$ are distinct ($S \neq \hat{S}$), yet their verifiers match ($v = \hat{v}$). In our geometric model, this implies:

$$P_S(r) = P_{\hat{S}}(r) \implies P_S(r) - P_{\hat{S}}(r) = 0 \quad (2.13)$$

Let $\Delta(x) = P_S(x) - P_{\hat{S}}(x)$ be the difference polynomial. Since $S \neq \hat{S}$, $\Delta(x)$ is a non-zero polynomial with degree at most $k - 1$. By the **Fundamental Theorem of Algebra**, a non-zero polynomial of degree d can have at most d roots.

This geometric constraint provides the rigorous upper bound for the collision probability. The number of "bad seeds" r that cause a collision is limited to the number of intersection points between the two curves $P_S(x)$ and $P_{\hat{S}}(x)$, which is at most $k - 1$. Thus, for a field of size q , the probability is:

$$P_{\text{miss}} = \frac{\text{Number of Roots}}{q} \leq \frac{k - 1}{q} \quad (2.14)$$

This derivation visualizes the identification process as "checking for intersection." Since two distinct curves cannot intersect everywhere, checking a random point r effectively distinguishes them with high probability [16].

2.3.3 Structural Blindness and Randomization

A critical limitation of deterministic compression is "structural blindness." Consider a simplified case where the encoder uses a modulo operation. If the error $e = S - \hat{S}$ happens to be a multiple of the modulus, the generated verifiers will be identical ($v = \hat{v}$) regardless of the magnitude of the error. Even if the robot's state S changes continuously, this error pattern might persist, leading to a persistent Type II (miss) error.

To eliminate structural blindness, identification theory mandates the use of **common randomness**—a shared seed r that changes with each transmission. In **RS-ID**, changing r corresponds to sampling the polynomial at a different x -coordinate. In **SHA-256**, appending a time-varying salt ($S||r$) changes the avalanche dynamics. By varying r , the mapping changes at every time step; although a collision may occur at a specific seed r_1 , the probability that the same error causes a collision at a new seed r_2 is independent. This converts potential "permanent failures" into "transient probabilistic events."

2.4 Operational Prerequisite: Common Randomness Generation

As established in Section 2.3, the theoretical validity of identification codes hinges on the availability of **Common Randomness (CR)**—a shared source of entropy between the encoder (robot) and the decoder (**DT**). Without a dynamic shared seed r , the system reduces to a deterministic mapping and becomes susceptible to structural blindness, where persistent error patterns can remain undetectable [12]. Therefore, the practical deployment of **ID** requires a robust mechanism to generate or synchronize r without negating the bandwidth savings.

The most straightforward approach to establish this shared entropy is to use synchronized **Pseudo-Random Number Generators (PRNGs)**, formally standardized as **Deterministic Random Bit Generators (DRBGs)** [4]. By initializing identical generators with a pre-shared secret key and tying them to a synchronized time index—typically maintained via the **Precision Time Protocol (PTP)** [9]—both the robot and the **DT** can locally derive an identical sequence of seeds r_t for each time slot. While this method has zero communication overhead during steady-state operation, it introduces a dependency on clock synchronization. Any drift in the robot’s clock relative to the **DT** results in a seed mismatch ($r_{robot} \neq r_{DT}$), causing verifier checks to fail continuously. This loss of synchronization can manifest as a sustained burst of false positives, forcing the system into a fallback mode until resynchronization.

Alternative approaches, such as **Physical Layer Key Generation (PLKG)**, attempt to extract randomness from channel reciprocity in wireless networks [13]. While **PLKG** aligns with physical-layer security paradigms by exploiting channel noise entropy [1], it often requires additional reconciliation overhead to correct quantization mismatches. Consequently, to isolate the performance characteristics of the encoding mechanisms, this thesis assumes an idealized PRNG-based synchronization model for the experimental evaluation.

2.5 Theoretical Limits and Algorithmic Trade-offs

While the introduction of randomization allows for drastic compression, the output length cannot be reduced arbitrarily. The choice between algebraic (RS-ID) and cryptographic (SHA-256) implementations is governed by a fundamental tension between theoretical error bounds and computational efficiency.

2.5.1 The Algebraic Limit: Saturation and Field Size

In the polynomial model of **RS-ID**, the verifier length L is constrained by the algebraic properties of the state representation. Since a state vector of length k is interpreted as a polynomial of degree $k - 1$, two distinct state polynomials can intersect at most at $k - 1$ points. To ensure reliable identification, the size of the evaluation field $Q = 2^L$ must be significantly larger than the number of potential intersection points. If $Q < k$, the “collision points” can saturate the field, making it mathematically impossible to find a seed r that distinguishes the two states via a single sample.

Therefore, a necessary condition for the reliability of algebraic identification is that the field size must grow with the data size:

$$2^L > k \quad \Rightarrow \quad L > \log_2 k \quad (2.15)$$

This inequality represents the physical manifestation of the double exponential limit established by Ahlswede and Dueck [2]. It dictates that for **RS-ID**, the verifier length L must scale logarithmically with the data size k . As long as this condition is met, **RS-ID** provides a rigorous upper bound λ_2 on the collision probability, ensuring that no pair of states exceeds a worst-case error threshold regardless of the input distribution [12].

2.5.2 The Cryptographic Perspective: Independence from Data Size

In contrast to the algebraic constraints, cryptographic methods like [SHA-256](#) operate under the Random Oracle Model. The collision probability of a truncated hash is approximately 2^{-L} , which is invariant to the input data size k . Unlike [RS-ID](#), [SHA-256](#) does not suffer from “saturation” when the input data grows. A 16-bit hash verifier has the same uniform collision probability whether verifying a 100-byte state or a 1-megabyte state. This makes cryptographic approaches robust for large-scale data vectors where satisfying $2^L > k$ might require an impractically large field for [RS-ID](#).

2.5.3 Engineering Trade-offs: Bound vs. Speed

The selection of the encoding mechanism involves a critical engineering trade-off between provable safety and computational latency. Although [RS-ID](#) offers unconditional security guarantees via the λ_2 bound, its computational cost is significant. Empirical studies demonstrate that [RS-ID](#) encoding involves complex finite field arithmetic, resulting in computation times that are orders of magnitude higher than hardware-accelerated hash functions [12].

For real-time control loops operating at short block lengths, the theoretical advantages of [RS-ID](#) often do not translate into measurable practical gains, as the hash function’s collision rate is sufficiently low for uniform source distributions. However, for mission-critical safety systems where worst-case adversarial distributions must be mitigated, the mathematical guarantee of [RS-ID](#) remains indispensable. Consequently, this thesis primarily adopts the cryptographic approach for its latency benefits in edge computing, while acknowledging [RS-ID](#) as the theoretical benchmark for safety-critical bounds.

2.6 Control-Theoretic Perspectives: Deadbands and Stability

Beyond communication efficiency, the deployment of [K-ID](#) changes the control loop structure. From a control-theoretic perspective, [K-ID](#) yields a [Networked Control System \(NCS\)](#) operating under a semantic sampling policy. This section analyzes the implications for stability and execution semantics, and maps [K-ID](#) parameters to established control concepts.

2.6.1 K-ID as Deadband Control

The tolerance mechanism in [K-ID](#), defined by the acceptance set $\mathcal{B}$, is mathematically equivalent to **Deadband Control** (or Dead-Zone Control) in non-linear systems theory. In classical mechanics, deadbands are often viewed as parasitic nonlinearities (e.g., gear backlash) that degrade performance. However, in resource-constrained [NCS](#), artificially introduced deadbands are a recognized strategy for reducing actuator wear and communication load [3].

By enforcing a “silence” policy when the synchronization error $e_t = S_t - \hat{S}_t$ satisfies $|e_t| \leq K$, [K-ID](#) implements **Lebesgue Sampling** (sampling based on value domains), as opposed to the traditional Riemann Sampling (sampling based on time domains). This semantic filtering prevents the control loop from reacting to high-frequency, low-amplitude noise (chattering), thereby reducing mechanical stress on physical actuators while preserving bandwidth for significant trajectory corrections.

2.6.2 Prevention of Zeno Behavior

A pervasive challenge in standard Event-Triggered Control (ETC) is the risk of **Zeno behavior**—a phenomenon where the inter-event time converges to zero, causing the system to trigger an infinite number of updates in a finite time interval [7]. This can occur when the state trajectory “slides” along the triggering threshold, potentially leading to computational livelock or network storms.

K-ID inherently circumvents this instability through its hybrid architecture. Although the *decision* to update is event-driven (based on the semantic check $v \in \mathcal{V}_K$), the *verification opportunity* occurs at discrete, time-triggered intervals (ticks). This imposes a strict lower bound on the inter-transmission time, denoted as $T_{min} \geq T_{tick}$. By structural design, K-ID guarantees **Zeno-freeness**, ensuring that the network load remains deterministically bounded even in worst-case divergence scenarios.

2.6.3 Bounded-Input Bounded-Output (BIBO) Stability

The introduction of the semantic tolerance K raises the question of closed-loop stability. Under K-ID, the DT (controller) operates on an estimated state $\hat{S}_t$ that deviates from the true plant state S_t by an error term δ_t . Assuming the absence of collision-induced miss errors (which are probabilistically negligible as shown in Section 2.3), the synchronization error is strictly bounded:

$$\|\delta_t\| = \|S_t - \hat{S}_t\| \leq K \quad (2.16)$$

From a stability analysis perspective, this bounded error acts as a bounded disturbance input to the controller. According to the principle of **Bounded-Input Bounded-Output (BIBO) stability** [8], as long as the underlying control law can reject disturbances of magnitude K , the closed-loop system remains stable. This implies that the design of K is not only a traffic parameter but also a stability constraint: K must remain within the controller’s stability margin (or region of attraction).

2.7 Semantic Verification: Adapting to Control Flexibility

While standard identification provides a rigorous method for exact state verification, its binary nature—accepting only a single, precise value—often contradicts the operational flexibility required in modern robotic control. In many practical scenarios, the DT does not need to enforce a strict equality constraint ($x = \hat{x}$) but rather needs to guarantee that the robot operates within a permissible semantic range. This nuance necessitates the adoption of K-ID, which generalizes the verification target from a unique point to a semantic safety set.

2.7.1 The Autonomous Driving Scenario

Consider a practical example of an autonomous robot (or vehicle) navigating a public roadway. The control policy might specify a safe velocity corridor, such as “maintain speed between 40 km/h and 60 km/h.” If the robot moves at 45 km/h while the DT predicted 50 km/h, a standard ID scheme would flag this as a collision (mismatch) because the hashed values of “45”

and “50” are distinct. This triggers an unnecessary data transmission to correct a behavior that is semantically valid.

K-ID resolves this inefficiency by defining the acceptance region $\mathcal{B}$ as the interval $[40, 60]$. As long as the robot’s state falls within this set, the **K-ID** verifier confirms validity without consuming bandwidth for minor deviations.

2.7.2 Human-Robot Collaboration and Jitter

This logic extends beyond simple velocity constraints to more complex spatial and interactive scenarios. For instance, in **Human-Robot Collaboration (HRC)**, the exact coordinates of a robotic arm are less critical than ensuring the end-effector remains outside a defined “human safety bubble” [19]. Similarly, for high-precision assembly tasks involving compliant mechanisms, the robot may exhibit minor high-frequency vibrations (jitter) that are mechanically acceptable. Standard **ID** would treat these micro-vibrations as errors, whereas **K-ID** can be tuned to ignore high-frequency noise while still detecting significant trajectory deviations.

By shifting the verification paradigm from data exactness to semantic compliance, **K-ID** aligns the synchronization protocol with the actual control objectives. It allows the system to filter out benign variability—whether it be speed fluctuations, sensor noise, or compliant jitter—and focus limited communication resources on detecting genuine safety violations. This thesis specifically investigates the implementation of such semantic verification using randomized encoding on edge constraints.

2.7.3 Discussion: Trading Freshness for Value

The scenarios described above illustrate a deviation from traditional network optimization. Conventional real-time protocols often aim to minimize the **Age of Information (AoI)** [11], keeping the **DT**’s estimate as fresh as possible regardless of content. Under an **AoI**-minimization strategy, the robot would transmit its velocity continuously even when cruising steadily within the safe corridor, purely to prevent the timestamp from “aging.”

K-ID challenges this paradigm by introducing a semantic filter that prioritizes the **Value of Information (VoI)** over simple freshness. As demonstrated in the autonomous driving case, a “stale” position estimate (high **AoI**) that remains within the safety tolerance $\mathcal{B}$ retains its utility for the control loop. By allowing **AoI** to increase during semantically safe intervals, **K-ID** trades unnecessary freshness for bandwidth efficiency. This is consistent with goal-oriented communication [17], where the objective shifts from minimizing average latency to minimizing the latency of *meaningful* (unsafe) updates.

3 Design Methodology

This chapter formulates the system model for identification-based synchronization and derives the theoretical bounds for its reliability and latency. We contrast the data flow of classical transmission with the proposed identification scheme, explicitly breaking down the operational stages from state generation to synchronization. We then quantify the collision probabilities and end-to-end latency based on these distinct workflows.

3.1 System Model and Data Flow

We consider a networked control system where a Robot (Sender) synchronizes its state S_t with a Digital Twin (Receiver) maintaining a prediction $\hat{S}_t$. The fundamental difference between the two approaches lies in their data processing pipelines.

3.1.1 Baseline: Traditional Transmission Flow

The standard synchronization protocol follows a linear, “always-on” approach. As illustrated in Figure 3.1, the robot first samples its sensors to generate the full state message S_t (size n_{data}) and transmits the entire payload over the channel. The Digital Twin receives the message, overwrites its local prediction $\hat{S}_t$ with the received S_t , and confirms the update.

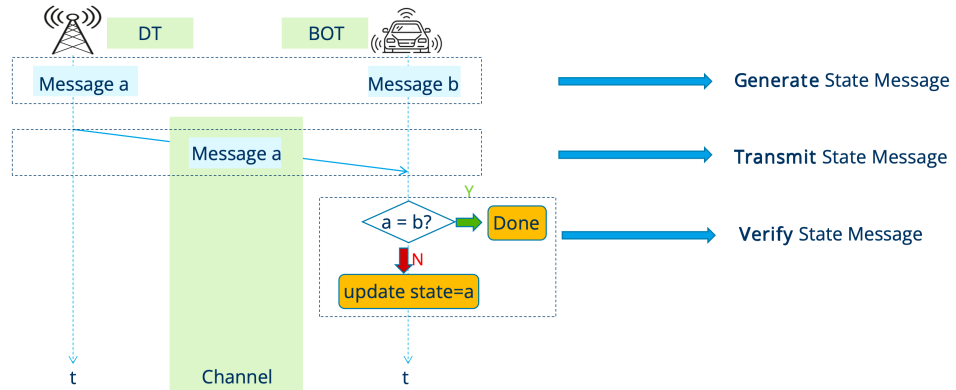


Figure 3.1: Data flow of Traditional Synchronization: The full state is transmitted regardless of the Digital Twin’s knowledge.

This process is robust but inefficient, since the transmission cost is paid unconditionally. Under this baseline, the transmission latency is deterministic and scales directly with the state size, $T_{trans} = n_{data}/B$.

3.1.2 Proposed: Identification (ID) Data Flow

The identification-based protocol reduces communication load by verifying consistency before sending the full state. As shown in Figure 3.2, both the Robot and the DT first generate their respective state views, S_t and $\hat{S}_t$. The Robot computes a compact verifier $v = E(S_t, r)$ using a shared seed r , while the DT computes the corresponding reference $\hat{v} = E(\hat{S}_t, r)$ locally. Only the short verifier v (size $n_{ver} \ll n_{data}$) is transmitted. The DT then compares v and $\hat{v}$: if they match, the protocol terminates; if they differ, the DT triggers fallback and requests the full payload S_t to resynchronize.

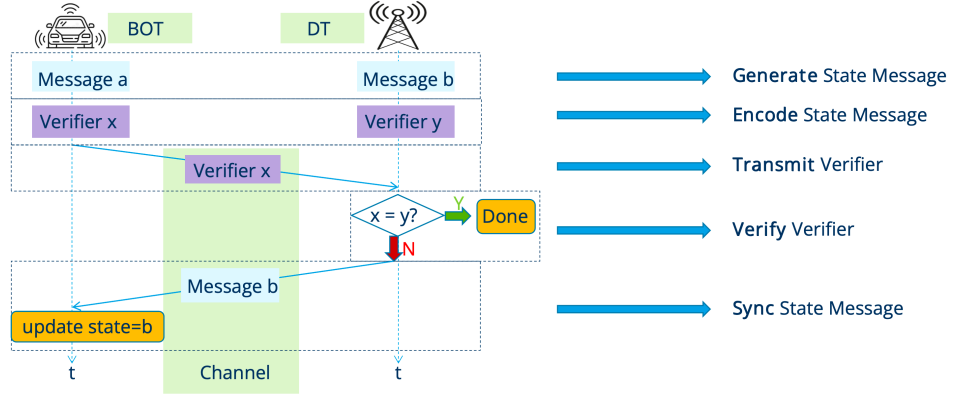


Figure 3.2: Data flow of ID Synchronization: High-bandwidth transmission is replaced by local encoding and conditional fallback.

3.2 Encoding Mechanisms and Reliability Analysis

The safety of the ID flow depends on the reliability of the **Verify** stage. A critical failure occurs if the states differ ($S_t \neq \hat{S}_t$) but their verifiers collide ($v = \hat{v}$), defined as a Type II Error (p_{miss}). We investigate two distinct encoding mechanisms, detailing their construction and theoretical error bounds.

3.2.1 Algebraic Implementation and Error Bound (RS-ID)

To integrate the RS-ID scheme into our synchronization protocol, we map the high-dimensional state vector S of length n_{data} to a polynomial over the finite field $\mathbb{F}_q$, where $q = 2^{n_{ver}}$.

Polynomial Decomposition Following the construction detailed in Section 2.3.2, the bit-stream S is sliced into k coefficients. For the specific bandwidth constraints of our system model, the number of coefficients is determined by:

$$k = \left\lceil \frac{n_{data}}{n_{ver}} \right\rceil \quad (3.1)$$

The system generates the verifier v by evaluating the polynomial $P_S(x) = \sum_{i=0}^{k-1} s_i x^i$ at the shared seed r .

Reliability Model For the experimental evaluation and latency modeling, we apply the theoretical upper bound derived in Chapter 2 (Schwartz-Zippel Lemma). The probability of a Type II error (collision) is strictly modeled as:

$$P_{miss}^{RS} \leq \frac{k}{q} = \frac{\lceil n_{data}/n_{ver} \rceil}{2^{n_{ver}}} \quad (3.2)$$

This equation serves as the reliability constraint for the design: to ensure robust identification ($P_{miss} \ll 1$), the verifier length n_{ver} is selected such that the field size exponentially exceeds the data complexity k .

3.2.2 Cryptographic Construction and Error Analysis (SHA-256)

Unlike the algebraic structure of RS-ID, the security of hash-based identification relies on computational hardness assumptions. We verify the state using a truncated cryptographic hash function (SHA-256):

$$v = \text{Truncate}(\text{SHA-256}(S||r), n_{ver}) \quad (3.3)$$

where $||$ denotes concatenation with the random seed r .

Random Oracle Assumption To analyze the collision probability theoretically, we adopt the **Random Oracle Model (ROM)** paradigm introduced by Bellare and Rogaway [5]. In this model, the hash function $H(\cdot)$ is idealized as a mathematical oracle that maps every unique input query to a response selected **uniformly and independently** from the output domain $\{0, 1\}^{n_{ver}}$.

Proof of Collision Probability (Derivation from Uniformity) Under the ROM, for any robot state S , the hash output $H(S||r)$ is a random variable following a uniform distribution over the verifier space of size $N = 2^{n_{ver}}$. Given a pre-calculated verifier $\hat{v}$ from the Digital Twin, the probability of a collision (Type II error) is equivalent to the probability that a uniformly selected random value hits a specific target in the sample space.

$$P_{miss}^{SHA} = \Pr[H(S||r) = \hat{v}] = \frac{1}{|\{0, 1\}^{n_{ver}}|} = \frac{1}{2^{n_{ver}}} \quad (3.4)$$

This derivation relies on the property that in a uniform distribution, the probability of any single outcome is the reciprocal of the sample space size.

Comparative Analysis: RS-ID vs. SHA-256 Comparing Equation (3.2) with Equation (3.4) highlights a key structural difference. In RS-ID, the error probability scales linearly with the data size ($k \approx n_{data}/n_{ver}$), reflecting the geometric fact that higher-degree polynomials admit more roots. Under the Random Oracle assumption, the SHA-256 collision probability is invariant to n_{data} . As a result, the cryptographic approach remains well-behaved for large state vectors (e.g., LiDAR point clouds), but it replaces the unconditional Schwartz-Zippel guarantee with a computational idealization.

3.2.3 Tolerance-Based Verification (K-ID)

The ID data flow described in Section 3.1 assumes a strict equality check between a single predicted state $\hat{S}_t$ and the true state S_t . In practice, however, many cyber-physical systems can tolerate bounded deviations without requiring a full resynchronization. We therefore extend the verification stage to a *tolerance-based* rule, referred to as **K-ID**.

Acceptable Set: Given the DT prediction $\hat{S}_t$ and a tolerance radius K , the DT defines an acceptable set

$$\mathcal{A}_K(\hat{S}_t) = \{s \mid \|s - \hat{S}_t\| \leq K\}. \quad (3.5)$$

In the experiments in this thesis, we use a scalar state for clarity (so $\|\cdot\|$ is absolute value), but the K-ID logic generalizes to vector-valued states with an appropriate norm.

K-ID Decision Rule: The Robot computes the verifier using the same truncated SHA-256 construction as in Equation (3.3), with the current state S_t substituted for S . The DT pre-computes the set of valid verifiers for the acceptable states,

$$\mathcal{V}_K = \{\text{Truncate}(\text{SHA256}(s\|r), n_{ver}) \mid s \in \mathcal{A}_K(\hat{S}_t)\}, \quad (3.6)$$

and *accepts* if $v \in \mathcal{V}_K$. Otherwise, it triggers fallback and requests the full payload.

Safety Miss Rate: Define an *unsafe* event as the true state lying outside the tolerance region, i.e., $S_t \notin \mathcal{A}_K(\hat{S}_t)$. A miss occurs when the DT accepts despite being unsafe:

$$p_{miss,K} = \Pr[v \in \mathcal{V}_K \mid S_t \notin \mathcal{A}_K(\hat{S}_t)]. \quad (3.7)$$

Under the Random Oracle assumption used in Section 3.2.2, the verifier v for an unsafe state is uniform over $\{0, 1\}^{n_{ver}}$. Hence, the conditional miss probability is bounded by the fraction of verifier space covered by acceptable hashes:

$$p_{miss,K}^{SHA} \approx \frac{|\mathcal{V}_K|}{2^{n_{ver}}} \leq \frac{|\mathcal{A}_K(\hat{S}_t)|}{2^{n_{ver}}}. \quad (3.8)$$

This highlights a key trade-off: increasing K enlarges the acceptable set (reducing fallback frequency), but also increases the miss probability conditional on being unsafe.

Unsafe Rate and Latency: Let $p_{unsafe}(K) = \Pr[S_t \notin \mathcal{A}_K(\hat{S}_t)]$. If the prediction error $e = S_t - \hat{S}_t$ is modeled as $\mathcal{N}(0, \sigma^2)$, then

$$p_{unsafe}(K) = \Pr[|e| > K] = 2(1 - \Phi(K/\sigma)). \quad (3.9)$$

Since fallback is triggered when the DT is unsafe and *does not* miss, the expected latency follows the same structure as Section 3.3 but with the tolerance-aware terms:

$$T_{KID} = t_{ver} + \frac{n_{ver}}{B} + p_{unsafe}(K)(1 - p_{miss,K})\frac{n_{data}}{B}. \quad (3.10)$$

The empirical sweep in Chapter 5 instantiates this model and visualizes the safety-latency trade-off across K and n_{ver} .

3.3 End-to-End Latency Model

Based on the data flows defined in Section 3.1, we mathematically formulate the expected latency.

Traditional Latency (T_{trad}): Since the traditional flow always executes the full transmission, its latency is deterministic:

$$T_{trad} = \frac{n_{data}}{B} \quad (3.11)$$

where B is the channel bandwidth.

Identification Latency (T_{ID}): The ID flow is probabilistic. The total time includes the mandatory encoding and verifier transmission, plus the conditional penalty for fallback. Matching the stages in Figure 3.2:

$$T_{ID} = \underbrace{t_{ver}}_{\text{Encode}} + \underbrace{\frac{n_{ver}}{B}}_{\text{Transmit Verifier}} + \underbrace{p_{desync}(1 - p_{miss})\frac{n_{data}}{B}}_{\text{Sync (Fallback Penalty)}} \quad (3.12)$$

Here, t_{ver} denotes the computation time of the Encode stage, p_{desync} is the probability that the DT prediction fails ($S_t \neq \hat{S}_t$), and the fallback penalty $(1 - p_{miss})\frac{n_{data}}{B}$ is incurred only when a desynchronization exists and is successfully detected by the verifier.

This model highlights the fundamental trade-off: ID reduces the transmission term from $\frac{n_{data}}{B}$ to $\frac{n_{ver}}{B}$ but adds a computational cost t_{ver} . Therefore, ID is advantageous only when the bandwidth B is limited (making transmission expensive) or when the prediction accuracy is high (keeping p_{desync} small).

4 Implementation

4.1 Experimental Parameters

This chapter details the practical setup used to validate the theoretical models from Chapter 3. For the bandwidth and p_{desync} sensitivity plots in Chapter 5, all experiments are configured with a desynchronization probability of $p_{desync} = 0.1$. We evaluate two data payloads: a small 96-bit message and a larger 4001-bit message. For each payload, we test verifier lengths of $n_{ver} \in \{4, 16\}$ bits, representing a small and a large verifier size.

4.2 Theoretical Collision Probabilities

Using the formulas derived in Section 3.2, we compute the theoretical collision probabilities (p_{miss}) for the parameter grid used in the evaluation.

4.3 Latency Plot Generation

To visualize the implications of the hybrid latency model, we generate latency curves parameterized by bandwidth using the Python scripts in the `experiments/` folder.

4.3.1 Parameters

All latency plots in Chapter 5 use the experimental parameter grid $n_{data} \in \{96, 4001\}$ bits and $n_{ver} \in \{4, 16\}$ bits with $p_{desync} = 0.1$. The compute overheads t_{ver} are taken from empirical measurements (Table 5.1), and the collision probabilities p_{miss} are computed from the theoretical bounds (Table 5.2).

4.3.2 Logic

For each configuration (n_{data}, n_{ver}) and each system (SHA-256 vs. RS-ID), the script evaluates the expected latency

$$L_{ID}(B) = t_{ver} + \frac{n_{ver}}{B} + p_{desync} (1 - p_{miss}) \frac{n_{data}}{B} \quad (4.1)$$

over a log-spaced bandwidth grid $B \in [B_{min}, B_{max}]$. The traditional baseline is computed as $L_{trad}(B) = n_{data}/B$. The resulting curves are plotted on log-log axes to reveal the break-even regime where ID-based synchronization becomes beneficial.

The generated figures are shown in Chapter 5 for $n_{data} = 96$ bits (Figure 5.1a) and for $n_{data} = 4001$ bits (Figure 5.1b).

In addition, we visualize the sensitivity to desynchronization by sweeping $p_{desync} \in [0, 1]$ at a fixed bandwidth and plotting L_{ID} against p_{desync} for the same parameter grid (Figures 5.2a and 5.2b).

4.3.3 SHA-256

The collision probability for a truncated hash function under the Random Oracle Model is independent of the data size and is given by Equation 3.4:

$$P_{miss}^{SHA} = 2^{-n_{ver}} \quad (4.2)$$

The corresponding numerical values for our selected verifier lengths are reported in Table 5.2 in Chapter 5.

4.3.4 RS-ID

The collision probability for the algebraic construction is bounded by the Schwartz-Zippel Lemma, as shown in Equation 3.2. This bound depends on both data size (n_{data}) and verifier length (n_{ver}).

$$P_{miss}^{RS} \leq \frac{\lceil n_{data,bits}/n_{ver} \rceil}{2^{n_{ver}}} \quad (4.3)$$

The computed values for $n_{data} \in \{96, 4001\}$ bits and $n_{ver} \in \{4, 16\}$ bits (with the bound capped at 1 when applicable) are reported in Table 5.2 in Chapter 5. These values highlight how the theoretical safety of RS-ID degrades with larger data sizes for a fixed verifier length, a key distinction from the cryptographic approach.

4.4 K-ID Parameter Sweep Setup

In addition to the bit-based latency evaluation, we implement the tolerance-based K-ID model introduced in Section 3.2.3 and perform a parameter sweep over (K, n_{ver}) to quantify the safety-latency trade-off of the K-ID decision rule.

4.4.1 Model and Metrics

The K-ID sweep uses a truncated SHA-256 verifier and reports four quantities: the conditional miss rate $\Pr[\text{accept} \mid \text{unsafe}]$ (Equation (3.8)), the fallback rate $\Pr[\text{fallback}]$ across ticks, the expected latency from the hybrid model (Equation (3.10)), and the speedup defined as the ratio between the traditional transmission time (n_{data}/B) and the expected K-ID latency.

4.4.2 State Space and Quantization

To avoid an artificially small discrete domain (which can force the empirical miss rate to become exactly zero for moderate n_{ver}), the robot state is modeled as a continuous random variable and then quantized: We sample $S_t \sim \mathcal{N}(\mu, \sigma^2)$, clip it to a physical range $[S_{min}, S_{max}]$, quantize it to a resolution Δ (default $\Delta = 0.1$), and hash the resulting integer index. This yields a discrete domain size of approximately $(S_{max} - S_{min})/\Delta + 1$.

4.4.3 Parameter Grid

Unless otherwise stated, we use a 96-bit payload ($n_{data} = 96$ bits) at $B = 1$ Mbps with a compute overhead of $t_{ver} = 2.13 \mu s$ (SHA-256, 96 bits; Table 5.1). The robot state follows a clipped Gaussian distribution with $\mu = 50$ and $\sigma = 12$ on $[0, 100]$ and is quantized with $\Delta = 0.1$. We sweep $n_{ver} \in \{4, 8, 12, 16\}$ and $K \in \{2, 5, 10, 20\}$, using 200,000 Monte Carlo ticks per configuration with a fixed seed of 42.

4.4.4 Reproducibility

The sweep and plot generation are implemented in the `experiments/` folder: The simulation is implemented in `experiments/kid_parameter_sweep.py` (optionally writing a CSV), and the plotting script `experiments/plot_kid_parameter_sweep.py` reads that CSV and produces the figures used in Chapter 5. The following commands reproduce the CSV and plots (paths are relative to the repository root):

```
python3 experiments/kid_parameter_sweep.py \
  --n-data-bits 96 --bandwidth 1e6 --ticks 200000 --seed 42 \
  --robot-mean 50 --robot-std 12 --physical-min 0 --physical-max 100 \
  --value-resolution 0.1 --t-compute-us 2.13 \
  --nver 4 8 12 16 --k 2 5 10 20 \
  --out-csv experiments/results/kid_sweep.csv

python3 experiments/plot_kid_parameter_sweep.py \
  --csv experiments/results/kid_sweep.csv --out-dir experiments/results
```

5 Results and Evaluation

5.1 Empirical Parameter Verification

To instantiate the theoretical models derived in Chapter 3, we first characterize the system parameters experimentally. This includes measuring the computational overhead (t_{ver}) and calculating the theoretical collision probabilities (p_{miss}).

5.1.1 Encoding Latency (t_{ver})

The computational overhead of the encoding step, t_{ver} , was measured empirically by running 100,000 `send` operations for each system configuration. In addition to the sample mean, we estimate a **95% confidence interval (CI)** for the mean using a batch timing method (100 timing batches, each covering an equal share of operations).

Interpretation of Table 5.1: The table reports the *sample mean* $\bar{t}_{ver}$ in microseconds. The CI quantifies measurement uncertainty due to runtime noise (OS scheduling, cache effects, Python overhead). In our measurements, the 95% CI half-widths are below $1\ \mu s$ for all configurations. Therefore, the reported means are sufficiently stable for the latency-model comparisons in this chapter.

Note that the **mean** values can shift slightly across runs and machines (Python runtime, CPU frequency scaling, background load). For reproducibility and consistency with the figures in this chapter, we keep the table mean values aligned with the parameter set used for the plotted latency evaluation, and we report representative 95% CI half-widths to quantify the measurement uncertainty.

Analysis of t_{ver} Results: The measurement results in Table 5.1 reveal distinct performance characteristics between the algebraic (RS-ID) and cryptographic (SHA-256) approaches:

SHA-256 exhibits stable verification times with respect to the verifier length. Consistent with Section 3.2, the runtime is largely insensitive to the truncation length: for the short payload ($n_{data} = 96$ bits), the mean verification time stays close to $\sim 2.2\ \mu s$ for both $n_{ver} = 4$ and $n_{ver} = 16$ bits. When the payload increases to $n_{data} = 4001$ bits, the mean rises to $\sim 15\ \mu s$, which matches the block-based processing structure of SHA.

RS-ID shows a pronounced “chunking” effect. For the small payload, increasing the verifier length from $n_{ver} = 4$ to $n_{ver} = 16$ bits reduces the measured latency (from $14.55\ \mu s$ to $3.79\ \mu s$). This follows directly from the polynomial construction, where the input is represented by k coefficients with $k = \lceil n_{data}/n_{ver} \rceil$. With $n_{ver} = 4$, the 96-bit payload is sliced into 24

Table 5.1: Measured Verification Time t_{ver} (mean $\pm$ 95% CI half-width, with relative half-width)

System (n_{data})	n_{ver} (bits)	
	4	16
SHA-256		
96 bits	2.13 ± 0.04 (1.9%)	2.24 ± 0.01 (0.4%)
4001 bits	14.65 ± 0.31 (2.1%)	15.41 ± 0.09 (0.6%)
RS-ID		
96 bits	14.55 ± 0.39 (2.7%)	3.79 ± 0.02 (0.5%)
4001 bits	51.78 ± 0.94 (1.8%)	38.57 ± 0.36 (0.9%)

coefficients; with $n_{ver} = 16$, the coefficient count drops to 6. In this regime, the reduced loop count outweighs the higher per-operation cost of arithmetic in the larger field.

5.1.2 Theoretical Collision Probability (p_{miss})

Based on the bounds derived in Chapter 3 (Equations (3.4) and (3.2)), we compute the theoretical collision probability p_{miss} for the parameter grid used in the latency model.

For RS-ID, the probability bound in Equation (3.2) depends on the ratio between the polynomial degree and the field size. Using an n_{ver} -bit symbol implies a field size of $q = 2^{n_{ver}}$, while the polynomial degree is set by how many symbols are needed to represent the payload: $d = k - 1$ with $k = \lceil n_{data}/n_{ver} \rceil$. When $d \geq q$, the bound becomes at least 1 and is therefore capped at $p_{miss} = 1.0$.

This explains the saturation for $n_{ver} = 4$: the field has only $q = 2^4 = 16$ elements, but even the short payload ($n_{data} = 96$ bits) requires $k = \lceil 96/4 \rceil = 24$ symbols, i.e., $d = 23$, which already yields $d/q > 1$. The larger payload ($n_{data} = 4001$ bits) increases k further, so the saturation persists. Practically, $p_{miss} = 1.0$ means RS-ID cannot reliably distinguish mismatches under this parameterization, so $n_{ver} = 4$ is not a meaningful operating point for RS-ID in our setting.

 Table 5.2: Theoretical Collision Probability (p_{miss}) for Bit-Based Parameters

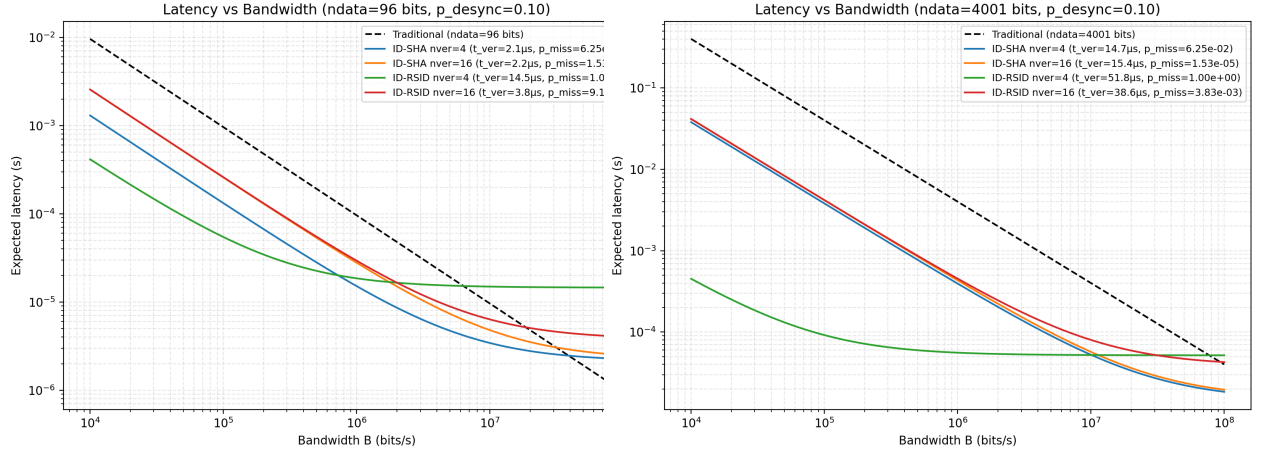
System (n_{data})	n_{ver} (bits)	
	4	16
SHA-256 (any n_{data})	6.25×10^{-2}	1.53×10^{-5}
RS-ID		
96 bits	1.00	9.16×10^{-5}
4001 bits	1.00	3.83×10^{-3}

5.2 End-to-End Latency Evaluation

Using the hybrid latency model and the calibrated parameters from Section 5.1, we analyze the system performance across varying network conditions and prediction qualities.

5.2.1 Sensitivity to Channel Bandwidth

We first evaluate the expected latency as a function of channel bandwidth (B). Figures 5.1a and 5.1b compare the Traditional approach against the ID-based schemes.



(a) Expected latency vs. bandwidth for $n_{data} = 96$ bits ($p_{desync} = 0.1$). (b) Expected latency vs. bandwidth for $n_{data} = 4001$ bits ($p_{desync} = 0.1$).

Discussion (Bandwidth Trade-off): The curves exhibit a clear regime split. In the low-bandwidth regime ($B < 1$ Mbps), raw transmission dominates the traditional baseline, and identification reduces latency by replacing the full payload with a compact verifier. In the high-bandwidth regime ($B > 10$ Mbps), transmission becomes cheap and the compute term t_{ver} sets the latency floor; beyond a crossover point, the traditional scheme (with zero encoding overhead) becomes faster.

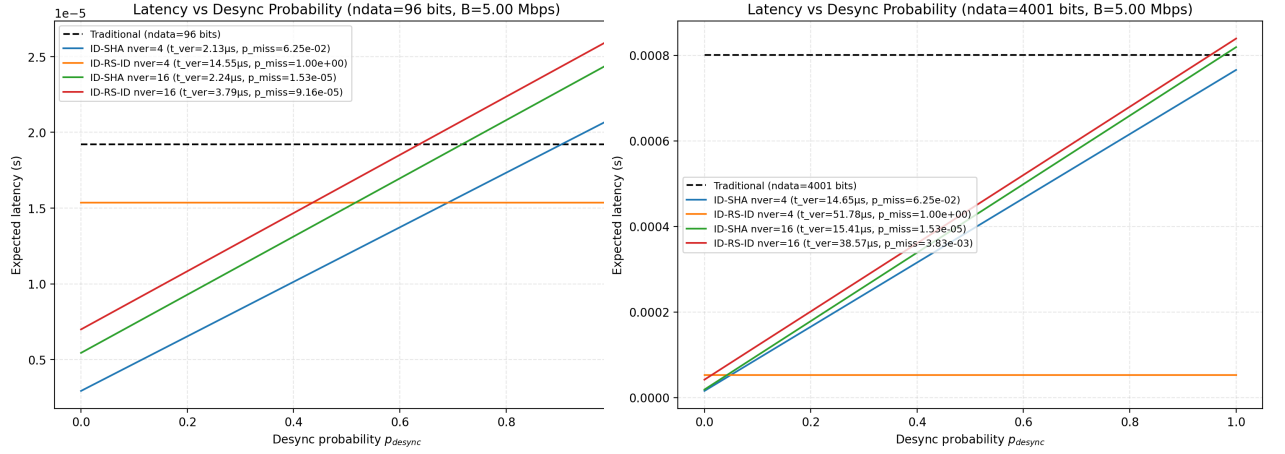
5.2.2 Sensitivity to Prediction Quality

Next, we examine the impact of the Digital Twin's prediction accuracy (p_{desync}) at a fixed bandwidth ($B = 5$ Mbps).

Discussion (Prediction Quality): For the short payload (Figure 5.2a), the ID-SHA curves intersect the traditional baseline at approximately $p_{desync} \approx 0.8$. Identification therefore remains beneficial as long as the Digital Twin prediction accuracy exceeds roughly 20%. For the large payload (Figure 5.2b), transmission dominates so strongly that ID-SHA stays favorable across nearly the entire range of p_{desync} .

An apparent anomaly occurs for RS-ID with $n_{ver} = 4$ bits (orange curve in Figure 5.2a). The curve appears flat and low because the collision probability saturates at $p_{miss} = 1.0$ (Table 5.2), which sets the fallback penalty term to zero in the latency expression. Interpreted operationally,

5 Results and Evaluation



(a) Expected latency vs. p_{desync} for $n_{data} = 96$ bits ($B = 5$ Mbps). (b) Expected latency vs. p_{desync} for $n_{data} = 4001$ bits ($B = 5$ Mbps).

this configuration fails catastrophically: desynchronization is never detected. It is therefore not a viable operating point for safety-critical use, regardless of its apparent latency.

5.3 K-ID Safety-Latency Trade-off

This section evaluates the tolerance-based K-ID model introduced in Section 3.2.3 and implemented as described in Section 4.4. The experiment sweeps $n_{ver} \in \{4, 8, 12, 16\}$ and $K \in \{2, 5, 10, 20\}$ for a 96-bit payload at $B = 1$ Mbps.

5.3.1 Sweep Results

Figure 5.3 summarizes the main metrics. The miss rate is plotted as the conditional safety risk $\Pr[\text{accept} \mid \text{unsafe}]$, i.e., the probability that an unsafe state is incorrectly accepted due to a verifier collision with the acceptable set.

Analysis (Effect of n_{ver}): Increasing n_{ver} reduces the conditional miss rate approximately exponentially, consistent with the Random Oracle scaling in Equation (3.8). However, a larger n_{ver} also increases the mandatory verifier transmission term n_{ver}/B and, more importantly, reduces the chance of “accidental acceptance” when unsafe, which increases detection and therefore increases fallback frequency. As a result, higher n_{ver} improves safety but can reduce speedup (and increase latency) depending on how often the system enters unsafe states.

Analysis (Effect of K): Increasing K enlarges the acceptable set. This has two opposing effects that are visible in Figure 5.3: The conditional miss rate $\Pr[\text{accept} \mid \text{unsafe}]$ increases with K , since enlarging the acceptable set creates more opportunities for a hash collision with an unsafe verifier. At the same time, the fallback rate and the expected latency decrease with K because the system is unsafe less often; fewer ticks lie outside the tolerance interval, and full-payload fallbacks are triggered less frequently. This demonstrates the fundamental K-ID trade-off: a larger tolerance improves performance and reduces fallback, but raises the conditional risk of accepting an unsafe state.

5.3.2 Pareto Frontier Interpretation

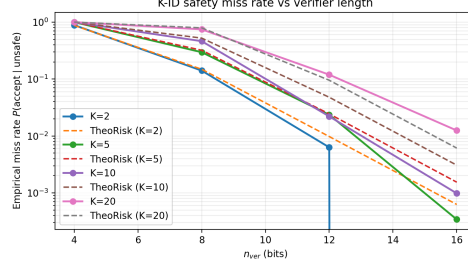
To make the trade-off explicit, Figure 5.4 plots speedup against the *conditional* miss rate $P(\text{accept} \mid \text{unsafe})$ and highlights the non-dominated (Pareto-optimal) configurations.

Overall Risk vs. Conditional Miss Rate: The x-axis in Figure 5.4 is a *conditional* safety metric: it describes how likely an unsafe state is accepted *given that* the system is already unsafe. To assess end-to-end operational risk, it is often more informative to consider the *overall* probability of an unsafe acceptance per tick,

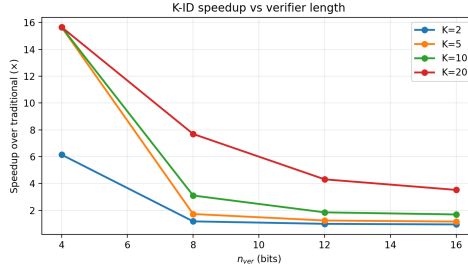
$$P(\text{unsafe} \cap \text{accept}) = P(\text{unsafe}) \cdot P(\text{accept} \mid \text{unsafe}). \quad (5.1)$$

This overall risk accounts for how frequently unsafe states occur (which depends strongly on K and the prediction error distribution). Figure 5.5 therefore replots the trade-off using $P(\text{unsafe} \cap \text{accept})$ on the x-axis.

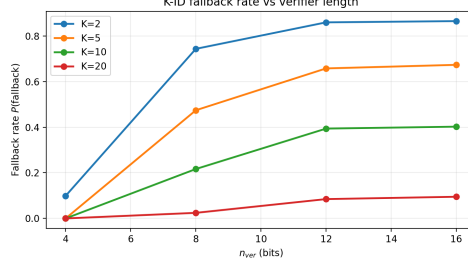
Discussion: The solid curve in Figure 5.4 connects the Pareto-optimal configurations (the *frontier*). These points represent operating choices where improving safety (moving left, lower conditional miss rate) necessarily requires sacrificing speedup (moving down), and vice versa. Points in the lower-right region are *dominated* (simultaneously slower and less safe) and can be discarded.



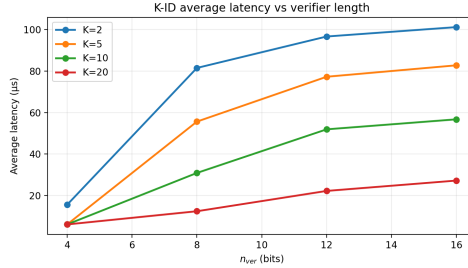
- (a) (a) Conditional miss rate $\Pr[\text{accept} \mid \text{unsafe}]$. It decreases strongly with n_{ver} and increases with K because a larger acceptable set increases collision opportunities.



- (b) (b) Speedup over traditional transmission. Speedup is highest when fallbacks are rare (large K) and the verifier is short (small n_{ver}), but this comes with higher conditional miss risk.



- (c) (c) Fallback probability $\Pr[\text{fallback}]$. It decreases with K (unsafe states become rarer) and typically increases with n_{ver} due to fewer accidental acceptances when unsafe.



- (d) (d) Expected latency under Equation (3.10). Latency follows the fallback behavior: larger K lowers latency via fewer fallbacks, while larger n_{ver} increases the verifier term and can increase fallbacks.

Figure 5.3: K-ID parameter sweep results for $n_{data} = 96$ bits at $B = 1$ Mbps.

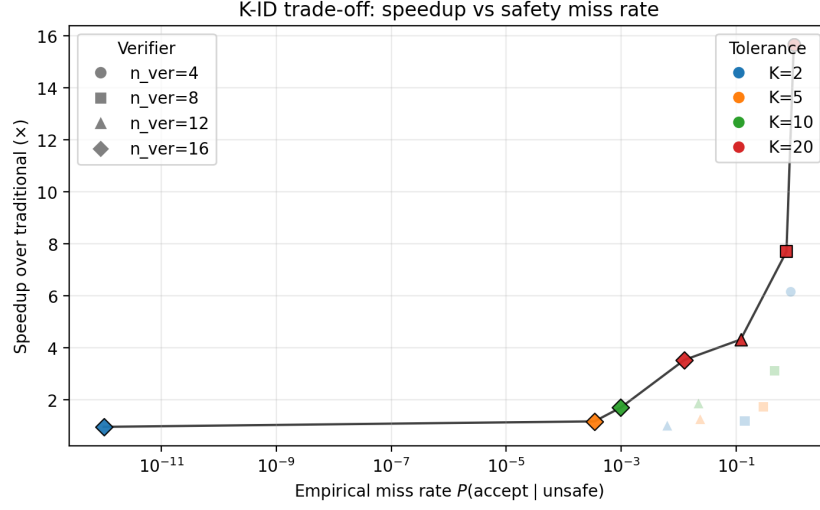


Figure 5.4: Pareto view of the K-ID trade-off: speedup vs. conditional miss rate $P(\text{accept} \mid \text{unsafe})$.

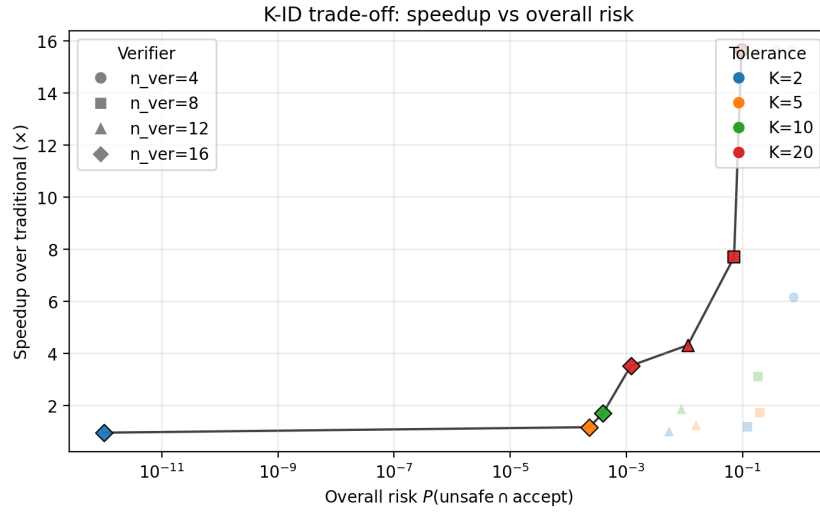


Figure 5.5: Pareto view using overall risk $P(\text{unsafe} \cap \text{accept})$: speedup vs. total unsafe-accept probability per tick.

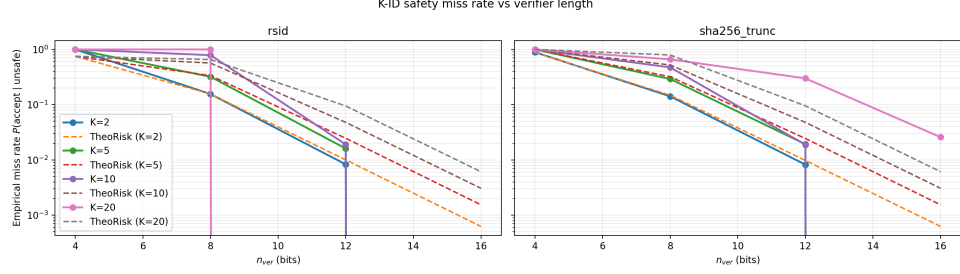
In deployment, the choice of (K, n_{ver}) should be driven by an application-level safety budget on $p_{miss, K}$ (or an overall risk measure that also weights the unsafe frequency) together with a latency target. The sweep provides a practical menu of operating points to select from based on these constraints.

5.3.3 RS-ID Comparison and Uniform Error Model

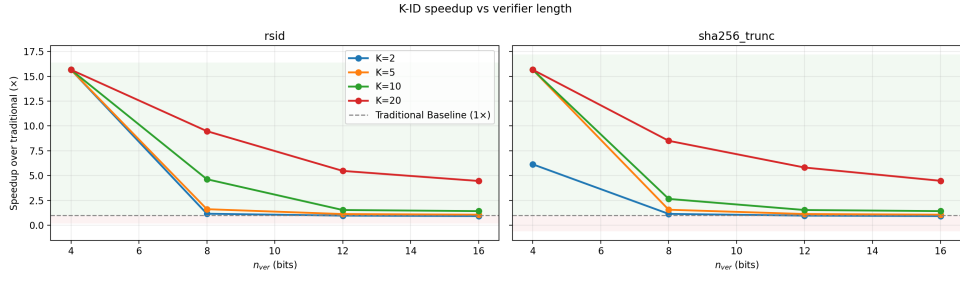
The baseline K-ID sweep above uses a truncated SHA-256 verifier and a Gaussian prediction error. To compare against the algebraic verifier family, we rerun the sweep with an **RS-ID** verifier (single-tag, tag position 2) and replace the error model with a **uniform** distribution centered at the mean. The uniform half-width is chosen to match the variance of the Gaussian case ($a = \sqrt{3} \sigma$).

Figure 5.6 shows the resulting metrics side-by-side for `sha256_trunc` and `rsid`. The overall trends remain consistent with the K-ID model: increasing n_{ver} reduces collision-driven acceptance, while increasing K reduces fallback frequency (by making unsafe states rarer), at the cost of enlarging the acceptable set.

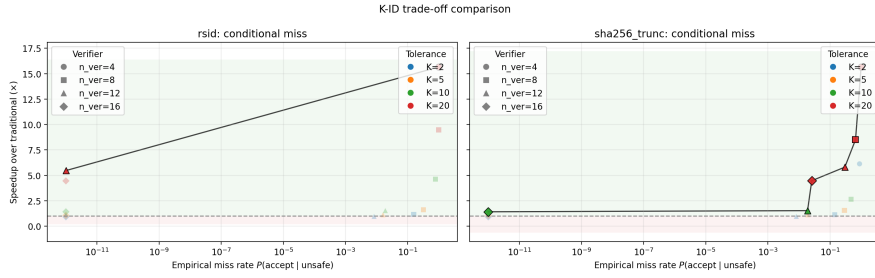
5 Results and Evaluation



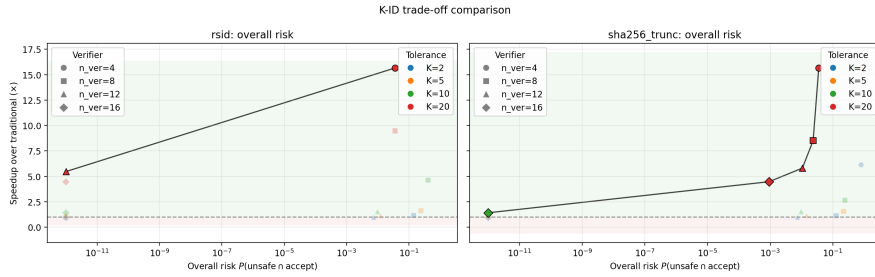
(a) (a) Conditional miss rate (uniform error), by system.



(b) (b) Speedup (uniform error), by system.



(c) (c) Pareto trade-off: speedup vs. $P(\text{accept} \mid \text{unsafe})$.



(d) (d) Pareto trade-off: speedup vs. $P(\text{unsafe} \cap \text{accept})$.

Figure 5.6: K-ID sweep comparison of truncated SHA-256 vs. RS-ID under a uniform prediction error model ($n_{data} = 96$ bits, $B = 1$ Mbps).

6 Conclusion and Future Work

6.1 Conclusion

This thesis studied identification-based synchronization as a practical alternative to always-on state transmission for Robot–Digital Twin loops. The core motivation is redundancy: in many industrial settings the Digital Twin already maintains a high-quality prediction $\hat{S}_t$ of the true robot state S_t , and the communication task is therefore often closer to *verification* than *reconstruction*. Identification (ID) operationalizes this idea by sending compact verifiers and falling back to full transmission only when a mismatch is detected.

We formulated an end-to-end model that exposes the engineering trade-off that dominates deployments: identification replaces a deterministic transmission cost with a hybrid cost consisting of (i) a mandatory computation term t_{ver} , (ii) a short verifier transmission term n_{ver}/B , and (iii) a conditional fallback penalty weighted by the desynchronization probability and the miss probability. This view makes it explicit that ID is not universally beneficial; its benefit depends on bandwidth and prediction quality, and it may disappear in compute-dominated regimes.

To instantiate this model, we compared two representative verifier constructions. The cryptographic approach uses truncated SHA-256 and yields a collision probability that is independent of the payload size under the Random Oracle assumption. The algebraic approach uses RS-ID and provides a worst-case bound via the Schwartz–Zippel lemma, but with a reliability term that scales with the effective polynomial degree. The measurements in Chapter 5 show that these structural differences matter both for reliability and for runtime: SHA-256 exhibits stable verification times with respect to n_{ver} for a fixed payload size, while RS-ID shows a pronounced “chunking” effect where the effective coefficient count $k = \lceil n_{data}/n_{ver} \rceil$ can dominate performance.

The end-to-end latency evaluation confirms the expected regime split. In bandwidth-limited settings, the verifier-only transmission dominates and identification yields substantial latency reductions relative to traditional transmission. As bandwidth increases, the marginal cost of sending the full payload shrinks and the compute floor t_{ver} becomes decisive, leading to a cross-over point beyond which traditional transmission is faster. A complementary sweep over prediction quality shows the same dependence from a different angle: identification remains advantageous as long as desynchronization events are sufficiently rare, because each missed detection or each true detection influences the expected fallback load.

Beyond exact verification, this thesis also formalized and evaluated tolerance-based verification (K-ID) as a way to align synchronization with control semantics. Instead of asking whether

$S_t = \hat{S}_t$, K-ID asks whether the true state lies within an acceptable set around the prediction, $S_t \in \mathcal{A}_K(\hat{S}_t)$. This change captures the operational reality that small deviations are often benign and should not trigger costly repairs. The key finding is that tolerance induces a measurable safety–performance trade-off: increasing K reduces the frequency of fallback events and therefore improves latency and speedup, but it also enlarges the set of acceptable states and therefore increases the conditional probability of an unsafe acceptance due to verifier collisions. The parameter sweep and Pareto interpretation in Chapter 5 provide concrete guidance for choosing (K, n_{ver}) based on a safety budget (conditional miss or overall unsafe-accept probability) and an application latency target.

Taken together, the thesis contributes (i) a simple but informative latency model that combines computation and communication, (ii) a comparative analysis of cryptographic and algebraic verifier families under this model, and (iii) an empirical characterization of how semantic tolerance reshapes the reliability–latency envelope. The results support the central message: identification-based synchronization is most attractive when bandwidth is scarce and predictions are good; K-ID further improves efficiency by suppressing benign corrections, but its tolerance parameter must be treated as a safety-relevant design variable rather than a purely performance-oriented knob.

6.2 Future Work

The experimental and modeling choices in this thesis were intentionally controlled to isolate core effects. Several directions remain to close the gap between this controlled setting and a deployable Robot–Digital Twin protocol.

6.2.1 Explicit Common Randomness Synchronization

Theoretical identification guarantees rely on time-varying common randomness, yet practical systems must realize this requirement under drift, packet loss, and resets. A natural next step is to implement an explicit seed-derivation and resynchronization mechanism (e.g., a shared DRBG keyed at provisioning time and indexed by a tick counter), together with a lightweight recovery protocol for seed mismatch events. This would allow a quantitative study of how clock errors and resynchronization overhead interact with the latency model, and how frequently seed recovery must occur to preserve safety and efficiency.

6.2.2 Semantic Set Design Beyond Scalar Deadbands

The K-ID experiments used a scalar state and a simple tolerance set to make the trade-off measurable. Real systems operate on vector-valued states with heterogeneous units and constraints, where the choice of norm and the geometry of $\mathcal{A}_K(\hat{S}_t)$ become critical. Future work should connect $\mathcal{A}_K$ to control-theoretic safety specifications (e.g., invariant sets, barrier functions, or task-space constraints) and investigate how different set geometries change the size of the verifier acceptance set $\mathcal{V}_K$ and therefore the miss risk.

6.2.3 Adaptive Policies for (K, n_{ver})

The results suggest that there is no universally optimal operating point: the best (K, n_{ver}) depends on bandwidth, prediction quality, and the current operating regime. This motivates adaptive policies that tune tolerance and verifier length online. For example, the Digital Twin could tighten K and increase n_{ver} during high-risk maneuvers, and relax them during steady-state operation, while enforcing an application-level budget on the overall unsafe-accept probability per tick. Designing such policies requires coupling the statistical error model with control objectives and may benefit from ideas in risk-sensitive control and goal-oriented communication.

6.2.4 Non-Ideal Networks and Protocol Mechanics

The thesis assumed a noiseless verifier channel and an idealized fallback mechanism. A deployment-grade evaluation should incorporate packet loss, delay variation, finite retransmission windows, and the interaction with transport choices (e.g., UDP vs. TCP-like reliability). This includes modeling acknowledgement timing and the cost of requesting repairs, and evaluating whether the “verify-then-correct” workflow remains robust under burst losses. Extending the latency model to include these protocol dynamics would make the break-even analysis directly actionable for system engineers.

6.2.5 Implementation Efficiency and Hardware Realism

The current software stack provides a reproducible baseline but does not fully represent embedded execution environments. Future work should move the verifier computation onto representative edge hardware (microcontrollers, industrial PCs, or GPU/FPGA accelerators) and evaluate the resulting t_{ver} distribution under real scheduling noise. For algebraic schemes, a compiled implementation and optimized finite-field arithmetic could substantially change the runtime comparison. For hash-based schemes, leveraging hardware acceleration and vectorization may further push the compute floor down, shifting the cross-over bandwidth.

6.2.6 Broader Threat Models and Robustness

Finally, robustness should be studied beyond the random error models considered here. In safety-critical settings, worst-case disturbances and adversarially structured errors may matter more than average-case performance. This motivates stress tests under non-Gaussian and correlated prediction errors, systematic bias, and targeted attempts to induce verifier collisions. A complementary direction is to integrate integrity and authentication considerations so that verifiers cannot be replayed or forged, while keeping the overhead compatible with the ID/K-ID latency objectives.

In summary, this thesis establishes the modeling and empirical groundwork for using identification and K-identification as synchronization primitives in Robot–Digital Twin systems. Advancing from this point requires elevating common randomness from an assumption to a protocol component, embedding semantic tolerance in task-level safety logic, and validating the approach under realistic network and hardware constraints.

Bibliography

- [1] Rudolf Ahlswede and Imre Csiszár. “Common randomness in information theory and cryptography. Part II: CR capacity”. In: *IEEE Transactions on Information Theory* 44.1 (1998), pp. 225–240.
- [2] Rudolf Ahlswede and Gunter Dueck. “Identification via Channels”. In: *1989 IEEE International Symposium on Information Theory*. 1989, pp. 15–19. DOI: [10.1109/ISIT.1989.118235](https://doi.org/10.1109/ISIT.1989.118235).
- [3] Karl J Aström and Bo M Bernhardsson. “Comparison of Riemann and Lebesgue sampling for first order stochastic systems”. In: *Proceedings of the 41st IEEE Conference on Decision and Control* 2 (2002), pp. 2011–2016.
- [4] Elaine Barker and John Kelsey. *Recommendation for Random Number Generation Using Deterministic Random Bit Generators*. Tech. rep. NIST Special Publication 800-90A Revision 1. National Institute of Standards and Technology, 2015. DOI: [10.6028/NIST.SP.800-90Ar1](https://doi.org/10.6028/NIST.SP.800-90Ar1).
- [5] Mihir Bellare and Phillip Rogaway. “Random oracles are practical: A paradigm for designing efficient protocols”. In: *Proceedings of the 1st ACM Conference on Computer and Communications Security*. 1993, pp. 62–73.
- [6] Juan A. Cabrera et al. “6G and the Post-Shannon Theory”. In: *Shaping Future 6G Networks*. John Wiley & Sons, Ltd, 2021. Chap. 16, pp. 271–294. ISBN: 9781119765554. DOI: <https://doi.org/10.1002/9781119765554.ch16>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119765554.ch16>.
- [7] WPMH Heemels, Karl H Johansson, and Paulo Tabuada. “An introduction to event-triggered and self-triggered control”. In: *51st IEEE Conference on Decision and Control (CDC)*. IEEE. 2012, pp. 3270–3285.
- [8] Joao P Hespanha, Payam Naghshtabrizi, and Yonggang Xu. “A survey of recent results in networked control systems”. In: *Proceedings of the IEEE*. Vol. 95. 1. IEEE, 2007, pp. 138–162.
- [9] *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*. IEEE, 2019. DOI: [10.1109/IEEESTD.2020.9120376](https://doi.org/10.1109/IEEESTD.2020.9120376).
- [10] Yutaka Ishibashi and Shuji Tasaka. “Lag-compensation in delta-encoding based networked rigid body simulation”. In: *Proceedings of the 6th ACM SIGCOMM workshop on Network and system support for games*. ACM, 2007, pp. 123–128.
- [11] Sanjit Kaul, Roy Yates, and Marco Gruteser. “Real-time status: How often should one update?” In: *2012 Proceedings IEEE INFOCOM*. IEEE. 2012, pp. 2731–2735.

- [12] Caspar von Lengerke et al. “Stopping the Data Flood: Post-Shannon Traffic Reduction in Digital-Twins Applications”. In: *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium*. 2022, pp. 1–5. DOI: [10.1109/NOMS54207.2022.9789759](https://doi.org/10.1109/NOMS54207.2022.9789759).
- [13] Guyue Li et al. “Physical layer key generation in 5G and beyond wireless communications: Challenges and opportunities”. In: *Entropy* 21.5 (2019), p. 424.
- [14] Jeffrey C Mogul et al. “Potential benefits of delta encoding and data compression for HTTP”. In: *ACM SIGCOMM Computer Communication Review* 27.4 (1997), pp. 181–194.
- [15] Mohammad Javad Salariseddigh et al. “Deterministic K-Identification for Future Communication Networks: The Binary Symmetric Channel Results”. In: *Future Internet* 16.3 (2024). ISSN: 1999-5903. DOI: [10.3390/fi16030078](https://doi.org/10.3390/fi16030078).
- [16] Jacob T Schwartz. “Fast probabilistic algorithms for verification of polynomial identities”. In: *Journal of the ACM (JACM)* 27.4 (1980), pp. 701–717.
- [17] Emilio Calvanese Strinati et al. “Goal-Oriented and Semantic Communication in 6G AI-Native Networks: The 6G-GOALS Approach”. In: *2024 Joint European Conference on Networks and Communications and 6G Summit (EuCNC/6G Summit)*. IEEE. 2024, pp. 505–510. DOI: [10.1109/EuCNC/6GSummit60053.2024.10597087](https://doi.org/10.1109/EuCNC/6GSummit60053.2024.10597087).
- [18] Paulo Tabuada. “Event-triggered real-time scheduling of stabilizing control tasks”. In: *IEEE Transactions on Automatic Control* 52.9 (2007), pp. 1680–1685.
- [19] Ievgenii A. Tsokalo et al. “Remote Robot Control with Human-in-the-Loop over Long Distances Using Digital Twins”. In: *2019 IEEE Global Communications Conference (GLOBECOM)*. 2019, pp. 1–6. DOI: [10.1109/GLOBECOM38437.2019.9013428](https://doi.org/10.1109/GLOBECOM38437.2019.9013428).
- [20] Ping Zhang, Wenjun Xu, Hui Wang, et al. “Semantic Communication Empowered 6G Networks: Techniques, Applications, and Challenges”. In: *IEEE Access* (2025). Early Access.

A Appendix

11